

Smart Kanban

An AI-powered Kanban board built with **React**, **Express**, **Tailwind CSS**, and **Google Gemini AI** that helps users organize tasks, prioritize work, and improve productivity through intelligent task analysis.



Features

Task Management

- Create, edit, and delete tasks
- Drag & drop tasks across columns
- Priority labels (High, Medium, Low)
- Search tasks instantly
- Filter tasks by priority
- Local storage persistence
- Responsive user interface

AI Workspace

-  Suggest task priority
-  Break complex tasks into actionable steps
-  Generate board summaries
-  Productivity coaching
-  AI-powered Kanban chat assistant

Dashboard

- Task statistics
 - Progress overview
 - Clean and modern UI
 - Responsive layout
-

Tech Stack

Frontend

- React

- Vite
- Tailwind CSS
- JavaScript (ES6+)

Backend

- Node.js
- Express.js

AI

- Google Gemini API

Deployment

- Frontend: Vercel
- Backend: Render

Project Structure

```
smart-kanban
├── public/
├── server/
│   └── server.js
├── src/
│   ├── assets/
│   ├── components/
│   ├── services/
│   ├── App.jsx
│   └── main.jsx
├── package.json
├── tailwind.config.js
└── README.md
```

AI Features

The AI assistant understands your current Kanban board and can:

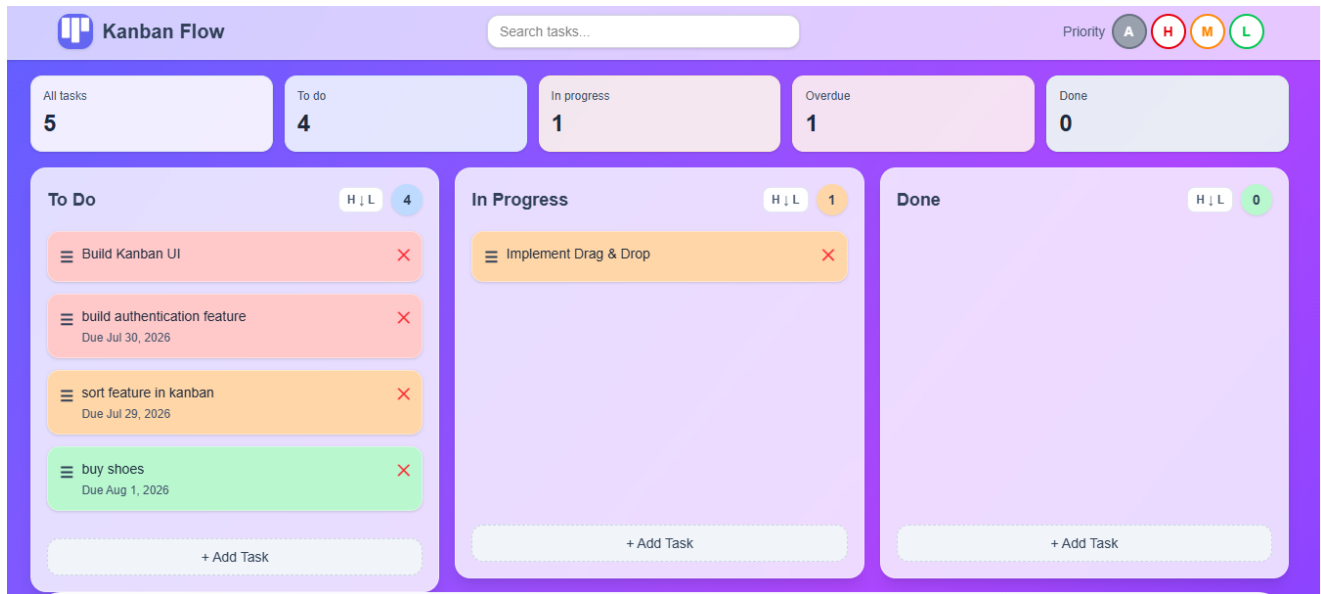
- Analyze task priority
- Break tasks into implementation steps
- Summarize project progress
- Identify risks
- Suggest next actions
- Answer project-related questions

- Provide productivity guidance

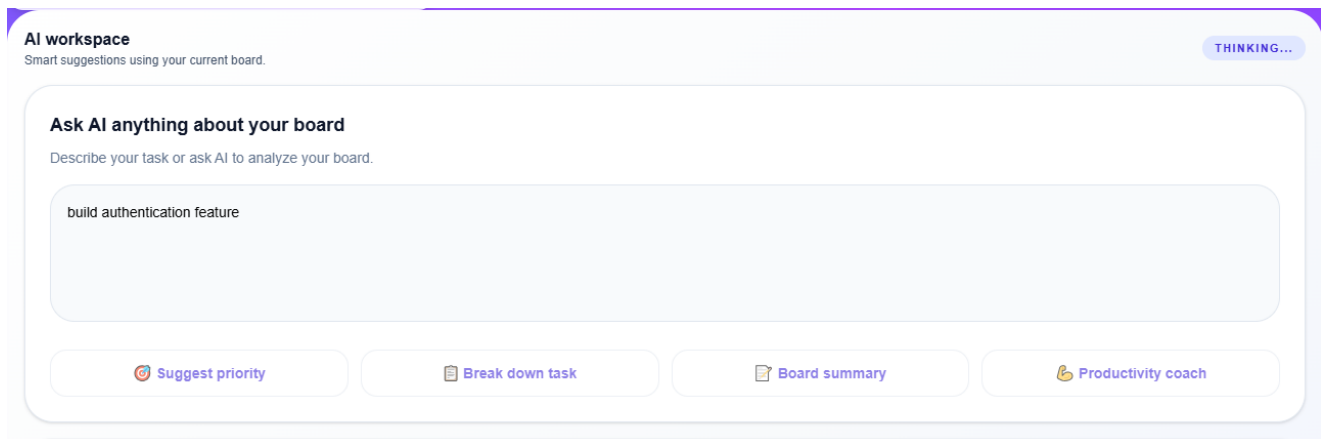
The backend securely communicates with the Google Gemini API, ensuring the API key is never exposed to the client.

Screenshots

Dashboard



AI Workspace



AI Chat Assistant

Search tasks...

Priority

A

H

M

L

AI chat assistant

Ask follow-up questions in a conversational chat format.

✖ PRIORITY

build authentication feature

✖ BREAKDOWN

build authentication feature

✖ COACH

build authentication feature

✖ SUMMARY

build authentication feature

Priority: High Reason:

This task is already designated as high priority on your board with an upcoming due date of 2026-07-30.

Task Breakdown 1. Set up API endpoints for user registration and login.

2. Design frontend login and signup form components.

3. Implement JWT or session-based state management.

Observation: The task build authentication feature is currently in your Todo list with high priority and a due date of 2026-07-30. You also have Implement Drag & Drop currently in progress. Advice: To maintain a smooth workflow, avoid switching contexts. Focus on completing your current in-progress task first before taking on another high-priority item. Keeping your work-in-progress limit low will help you deliver faster with fewer errors. Next Step: Finish Implement Drag & Drop and move it to Done, then move build authentication feature into inprogress.

Board Summary Progress:

- 0 of 5 tasks completed.
- 1 task in progress and 4 tasks in To Do. Risks:
- No completed tasks yet.
- High priority items are sitting in To Do with upcoming deadlines. Next Best Action:
- Finish Implement Drag & Drop to clear current in-progress work. Regarding "build authentication feature":
- Current Status: To Do
- Priority: High
- Recommendation: Assign this as the next task after clearing in-progress work to meet the 2026-07-30 due date.

Ask about your tasks...

Send

Board Summary

✖ SUMMARY

build authentication feature

Board Summary Progress:

- 0 of 5 tasks completed.
- 1 task in progress and 4 tasks in To Do. Risks:
- No completed tasks yet.
- High priority items are sitting in To Do with upcoming deadlines. Next Best Action:
- Finish Implement Drag & Drop to clear current in-progress work. Regarding "build authentication feature":
- Current Status: To Do
- Priority: High
- Recommendation: Assign this as the next task after clearing in-progress work to meet the 2026-07-30 due date.

Ask about your tasks...

Send

Installation

Clone the repository

```
git clone https://github.com/KeerthyHQ/smart-kanban.git
cd smart-kanban
```

Install dependencies

```
npm install
```

Start the frontend

```
npm run dev
```

Start the backend

```
npm run server
```

Environment Variables

Create a `.env` file inside the project root.

```
PORT=3001
GEMINI_API_KEY=YOUR_API_KEY
GEMINI_MODEL=gemini-flash-latest
```

How It Works

```
graph TD
    A[React Frontend] --> B[Express Backend]
    B --> C[Google Gemini API]
```

The frontend sends AI requests to the Express backend.

The backend securely calls the Gemini API and returns AI-generated responses.



Future Enhancements

- User Authentication
 - Cloud Database
 - Team Collaboration
 - Due Dates & Reminders
 - Activity History
 - Dark Mode
 - Analytics Dashboard
 - Export & Import Boards
-



Key Concepts Practiced

- React Components
 - React Hooks
 - State Management
 - Drag & Drop UI
 - REST APIs
 - Express.js
 - AI Integration
 - Prompt Engineering
 - Environment Variables
 - Responsive Design
 - Local Storage
 - Component Architecture
-



Author

Keerthika M

GitHub: [KeerthyHQ](#)

LinkedIn: [Keerthika M](#)



Support

If you found this project useful, consider giving it a ☆ on GitHub.